

is in a direction from M to N. Hence this current goes from N, through R, to M.

- 25** A 240 V supply is connected to an ideal transformer. The output of the transformer is connected to five 12 V, 24 W lamps connected in parallel. The lamps are operating normally.

What is the current drawn from the supply?

- A** 0.10 A **B** 0.50 A **C** 2.0 A **D** 10.0 A

Ans: B

The five lamps are all parallel to the secondary coil. Current through one lamp = $24/12 = 2.0$ A

Current through secondary coil, $I_s = 5 \times 2.0 = 10$ A

$$V_s I_s = V_p I_p,$$

$$12 \times 10 = 240 \times I_p$$

$$I_p = 0.50 \text{ A}$$

26 The graph shows an alternating current with a square waveform